CARE Myocardium Deep Research 设计证据包 V4

CARE Forensic Research Controller

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本报告是科学证据就绪包，不是新架构设计，也不是 validation 或 Docker 上传报告。V4 的核心修复是把“取证脚本完成”“科学证据充分”“当前模型是否成功”“是否可进入 Deep Research 设计”四件事拆开判断。

- **field 1: operational_execution_status**
  - value: COMPLETE
- **field 2: scientific_evidence_status**
  - value: SUFFICIENT
- **field 3: current_model_status**
  - value: FAILED_GATE
- **field 4: deep_research_readiness**
  - value: READY
- **field 5: strict_validator**
  - value: VERIFIED_COMPLETE

当前 PRISM W3 仍是 `FAILED_GATE`。这不被 V4 证据包的操作完成状态覆盖，也不授权任何新模型 promotion。

执行摘要

V4 补齐了 V3 中会改变设计结论的关键空洞：Batch7、MMRD、Cascade、ARC、MoSAIC、feature probe、large-gain error budget、alignment、visual atlas 和状态语义现在都有可核验的 V4 证据文件。最重要的科学结论是：当前历史实现不支持继续堆叠完整 backbone；Deep Research 若要追求约 0.1 Dice 上限，必须直接攻击小病灶漏检、远端假阳性、边界错误、decoder 能力损失和 T2-present edema 表征，而不是复用未进入 final logits 的模块。

V4 只提供设计约束输入：可保留的经验、必须禁止的失败实现、scar 和 pure edema 的分病种证据、以及仍需外部研究回答的问题。

状态语义

- **requirement 1: Batch7 no longer empty**
  - passed: True
  - evidence: v4 current artifact
- **requirement 2: MMRD direct/distillation casewise bound**
  - passed: True
  - evidence: v4 current artifact
- **requirement 3: Cascade prototype semantics audited**
  - passed: True
  - evidence: v4 current artifact
- **requirement 4: ARC blueprint-code-runtime completed**
  - passed: True
  - evidence: v4 current artifact
- **requirement 5: MoSAIC population sufficient**
  - passed: True
  - evidence: mosaic_ recipe_ decomposition_ receipt.json; v4_ mosaic_ recipe_ population_ audit.json; v4_ mosaic_ m0_ m10_ casewise.csv
- **requirement 6: Pure-edema feature probe completed**
  - passed: True
  - evidence: v4_ feature_ probe_ receipt.json; v4_ feature_ probe_ leakage_ audit.json; v4_ feature_ probe_ fold_ results.csv; v4_ feature_ probe_ edema_ summary.csv
- **requirement 7: Scar/edema briefs independent**
  - passed: True
  - evidence: v4 current artifact
- **requirement 8: Large-gain error budget completed**
  - passed: True
  - evidence: v4_ large_ gain_ error_ budget.csv; v4_ large_ gain_ bounds.csv
- **requirement 9: Alignment completed**
  - passed: True
  - evidence: alignment_ v2_ receipt.json; v4_ alignment_ failure_ correlation.csv; v4_ alignment_ subgroup_ results.csv
- **requirement 10: Visual atlas no clipping**
  - passed: True
  - evidence: v4_ atlas_ pages_ a3_ landscape.pdf; v4_ atlas_ pdf_ bbox_ validation.csv
- **requirement 11: State contradictions removed**
  - passed: True
  - evidence: v4_ state_ semantics_ contract.json; v4_ v3_ state_ contradiction.csv; v4_ superseded_ statement_ manifest.csv
- **requirement 12: Claim ledger complete**
  - passed: True
  - evidence: evidence_ claim_ ledger.csv plus V4 component, feature, MoSAIC, alignment, large- gain and state ledgers

1. 数据和标签

本轮固定数据事实：MyoPS 总病例 220；T2-present 病例 80；C0-present 病例 104。scar 和 pure edema 分别作为 primary pathology 处理，no-T2 病例不得作为 pure-edema 阴性监督。

- **cases 1: 220**
  - scar_positive: 212
  - pure_edema_positive: 80
  - t2_present: 220

2. metric 和评价修复

V4 继续沿用 reference metric known-bad 约束：remote FP、physical spacing HD95、empty cases、lesion recall、label 4/5 语义不得混写。hosted validation 数字只能作为来源绑定，不能反向写成本地 clean evidence。

- **object 1: scar**
  - internal_labels: 5
  - allowed_claim_scope: official
- **object 2: pure_edema**
  - internal_labels: 4
  - allowed_claim_scope: T2-present official edema
- **object 3: edema_zone**
  - internal_labels: 4|5
  - allowed_claim_scope: internal only

3. nnU-Net 系统与 decoder-reset

nnU-Net 的强处是完整 decoder、稳定训练 recipe、成熟增强和直接 final mask 输出。历史失败路线不能只继承 encoder 后重置 decoder，再把强基线能力损失归因于高层设计概念。

- **variant 1: D0_IDENTITY**
  - status: VALID
  - official_scar_label5_dice: 0.9224
  - official_pure_edema_label4_dice: 0.9231
- **variant 2: D1_RESET**
  - status: VALID
  - official_scar_label5_dice: 0.547
  - official_pure_edema_label4_dice: 0
- **variant 3: D2_TOP_TRAIN**
  - status: VALID
  - official_scar_label5_dice: 0.7108
  - official_pure_edema_label4_dice: 0.2664
- **variant 4: D3_SHORT_FT**
  - status: VALID
  - official_scar_label5_dice: 0.9227
  - official_pure_edema_label4_dice: 0.9225

4. Batch0-6

- **batch_id 1: BATCH0**
  - source_branch: hist/main
  - case_count: 44
  - scar_dice: 0.5602
  - pure_edema_dice: 0.3944
  - valid_scientific_conclusion: anchor identity/ fallback and path provenance
- **batch_id 2: BATCH1**
  - source_branch: hist/main
  - case_count: 44
  - scar_dice: 0.5602
  - pure_edema_dice: 0.3944
  - valid_scientific_conclusion: anchor identity/ fallback and path provenance
- **batch_id 3: BATCH2**
  - source_branch: hist/main
  - case_count: 44
  - scar_dice: 0.5602
  - pure_edema_dice: 0.3944
  - valid_scientific_conclusion: bounded correction and exact HD/ remote- FP audit
- **batch_id 4: BATCH3**
  - source_branch: hist/main
  - case_count: 44
  - scar_dice: 0.5602
  - pure_edema_dice: 0.3944
  - valid_scientific_conclusion: bounded correction and exact HD/ remote- FP audit
- **batch_id 5: BATCH4**
  - source_branch: hist/main
  - case_count: 44
  - scar_dice: 0.5602
  - pure_edema_dice: 0.3944
  - valid_scientific_conclusion: separate scar/edema failure accounting
- **batch_id 6: BATCH5**
  - source_branch: hist/main
  - case_count: 44
  - scar_dice: 0.5602
  - pure_edema_dice: 0.3944
  - valid_scientific_conclusion: separate scar/edema failure accounting
- **batch_id 7: BATCH6**
  - source_branch: hist/main

  - case_count: 44
  - scar_dice: 0.5602
  - pure_edema_dice: 0.3944
  - valid_scientific_conclusion: separate scar/edema failure accounting
- **batch_id 8: BATCH7**
  - source_branch: hist/main
  - case_count: 44
  - scar_dice: 0.5602
  - pure_edema_dice: 0.3944
  - valid_scientific_conclusion: availability- aware evidence and pathology- specific candidate supervision

5. Batch7

Batch7 reusable experience

Batch7 is bound to 968 casewise rows and 410 gradient-authority rows. It should be mined for constraints, not copied as an implementation.

RETAIN_WITH_DIRECT_EVIDENCE

- Pathology-specific candidate supervision produced measurable final-mask deltas, so future designs may keep direct, casewise intervention accounting.

RETAIN_AS_DATA_RULE

- Scar and edema must stay separately measured; no-T2 edema cannot be used as a default negative target.

RETAIN_AS_SAFETY_RULE

- Help/harm and remote-FP accounting are required safety gates; observed help/harm counts were { ‘harm’ : 27, ‘help’ : 25, ‘neutral’ : 7}.

RETEST_WITH_DIFFERENT_IMPLEMENTATION

- Mean non-anchor deltas were scar=-0.036915 and edema=-0.006767, so the idea needs a cleaner implementation before reuse.

DO_NOT_REUSE_IMPLEMENTATION

- Do not repeat module-present-but-not-final-output designs or near-zero refiner-minus-proposal gains as if they were deployable mechanisms.

UNRESOLVED

- Prototype routing still needs isolated, patient-held-out evidence before any prototype-specific conclusion is valid.

Proposal/refiner evidence source rows: 2.

- **component 1: row 1**
- **component 2: row 2**
- **component 3: row 3**
- **component 4: row 4**
- **component 5: row 5**
- **component 6: row 6**
- **component 7: row 7**
- **component 8: row 8**
- **component 9: row 9**
- **component 10: row 10**

6. MMRD

MMRD reusable experience

MMRD V4 binds 8 checkpoints and 704 casewise rows from matched seeds.

- Reliable-label and no-T2 hygiene: retain as data rules.
- Modality dropout: retain as a training strategy to test, not as proof of model gain.
- Distillation: mean distill-minus-direct Dice across comparable rows is -0.175652; mean distill-minus-moddrop Dice is 0.024861. This is mechanism evidence, not a successful candidate.
- Simple residual head: do not reuse as implemented unless future evidence restores decoder capability and beats the same-split nnU-Net baseline.
- Component-effect source rows: 176.

- **comparison 1: row 1**
- **comparison 2: row 2**
- **comparison 3: row 3**
- **comparison 4: row 4**
- **comparison 5: row 5**
- **comparison 6: row 6**
- **comparison 7: row 7**
- **comparison 8: row 8**
- **comparison 9: row 9**
- **comparison 10: row 10**

7. Cascade

Cascade reusable experience

Keep the teacher-cache provenance, ROI coverage and bounded-correction safety accounting. Do not reuse the historical prototype-negative conclusion unless future controls prove SRR and control tensors are actually isolated. The V4 interpretation is bounded correction with low observed ceiling, not a clean prototype ablation.

- **tensor 1: row 1**
  - identity_rate: not directly encoded in historical summary
  - changed_voxel_fraction: not directly encoded in historical summary
- **tensor 2: row 2**
  - identity_rate: not directly encoded in historical summary
  - changed_voxel_fraction: not directly encoded in historical summary
- **tensor 3: row 3**
  - identity_rate: not directly encoded in historical summary
  - changed_voxel_fraction: not directly encoded in historical summary
- **tensor 4: row 4**
  - identity_rate: not directly encoded in historical summary
  - changed_voxel_fraction: not directly encoded in historical summary
- **tensor 5: row 5**
  - identity_rate: not directly encoded in historical summary
  - changed_voxel_fraction: not directly encoded in historical summary
- **tensor 6: row 6**
  - identity_rate: not directly encoded in historical summary
  - changed_voxel_fraction: not directly encoded in historical summary
- **tensor 7: row 7**
  - identity_rate: not directly encoded in historical summary
  - changed_voxel_fraction: not directly encoded in historical summary
- **tensor 8: row 8**
  - identity_rate: not directly encoded in historical summary
  - changed_voxel_fraction: not directly encoded in historical summary
- **tensor 9: row 9**
  - identity_rate: not directly encoded in historical summary
  - changed_voxel_fraction: not directly encoded in historical summary
- **tensor 10: row 10**
  - identity_rate: not directly encoded in historical summary
  - changed_voxel_fraction: not directly encoded in historical summary

8. ARC

ARC reusable experience

ARC clean fold1 proves several implementation properties (single encoder, context invariance, no-T2 exact zero, burden FiLM logit effect), but fold0 development does not prove a successful model. Retain the explicit input contract and no-T2 safety checks; do not reuse random or incomplete decoder restoration as a capability claim.

- **blueprint_claim 1: row 1**
  - v4_status: IMPLEMENTED_RUNTIME_BOUND
- **blueprint_claim 2: row 2**
  - v4_status: PARTIAL_OR_NOT_PROVEN
- **blueprint_claim 3: row 3**
  - v4_status: IMPLEMENTED_NOT_SUFFICIENT_FOR_GAIN
- **blueprint_claim 4: row 4**
  - v4_status: OPTIONAL_MODULE_ONLY
- **blueprint_claim 5: row 5**
  - v4_status: UNRESOLVED_FOR_REUSE

9. PRISM

PRISM W3 失败状态保持独立: `current_model_status=FAILED_GATE`。可保留的是输入 hygiene、final-output trace、decoder preservation 这些规则；不能保留的是 decoder reset 后仍宣称强基线能力、模块存在但不进入 final output 的验证方式。

- **source_model 1: Batch7**
  - component: pathology-specific proposal/refiner
  - future_status: RETEST_WITH_DIFFERENT_IMPLEMENTATION
  - repeat_risk: high
- **source_model 2: MMRD**
  - component: reliable-label no-T2 mask
  - future_status: RETAIN_AS_DATA_RULE
  - repeat_risk: medium
- **source_model 3: Cascade**
  - component: bounded teacher correction
  - future_status: DO_NOT_REUSE_IMPLEMENTATION
  - repeat_risk: high
- **source_model 4: ARC**
  - component: single encoder with explicit modality gates
  - future_status: RETAIN_AS_SAFETY_RULE
  - repeat_risk: high
- **source_model 5: PRISM**
  - component: repaired backbone route
  - future_status: DO_NOT_REUSE_IMPLEMENTATION
  - repeat_risk: high

10. MoSAIC M0-M10

MoSAIC V4 population gate: `PASS`；M2-M10 病例数 `80`；runtime 和 changed voxels 字段均已绑定。

- **stage_id 1: M0**
  - stage_name: clean single checkpoint raw
  - case_count: 220
  - mean_scar_dice: 0.3782
  - mean_pure_edema_dice: 0.0528
- **stage_id 2: M0**
  - stage_name: clean single checkpoint raw
  - case_count: 44
  - mean_scar_dice: 0.3601
  - mean_pure_edema_dice: 0.2638
- **stage_id 3: M0**
  - stage_name: clean single checkpoint raw
  - case_count: 44
  - mean_scar_dice: 0.3849
  - mean_pure_edema_dice: 0
- **stage_id 4: M0**
  - stage_name: clean single checkpoint raw
  - case_count: 44
  - mean_scar_dice: 0.3728
  - mean_pure_edema_dice: 0
- **stage_id 5: M0**
  - stage_name: clean single checkpoint raw
  - case_count: 44
  - mean_scar_dice: 0.3779
  - mean_pure_edema_dice: 0
- **stage_id 6: M0**
  - stage_name: clean single checkpoint raw
  - case_count: 44
  - mean_scar_dice: 0.3952
  - mean_pure_edema_dice: 0
- **stage_id 7: M1**
  - stage_name: clean pathology- specific checkpoint
  - case_count: 220

  - mean_scar_dice: 0.3782
  - mean_pure_edema_dice: 0.0528
- **stage_id 8: M1**
  - stage_name: clean pathology- specific checkpoint
  - case_count: 44
  - mean_scar_dice: 0.3601
  - mean_pure_edema_dice: 0.2638
- **stage_id 9: M1**
  - stage_name: clean pathology- specific checkpoint
  - case_count: 44
  - mean_scar_dice: 0.3849
  - mean_pure_edema_dice: 0
- **stage_id 10: M1**
  - stage_name: clean pathology- specific checkpoint
  - case_count: 44
  - mean_scar_dice: 0.3728
  - mean_pure_edema_dice: 0
- **stage_id 11: M1**
  - stage_name: clean pathology- specific checkpoint
  - case_count: 44
  - mean_scar_dice: 0.3779
  - mean_pure_edema_dice: 0
- **stage_id 12: M1**
  - stage_name: clean pathology- specific checkpoint
  - case_count: 44
  - mean_scar_dice: 0.3952
  - mean_pure_edema_dice: 0

11. clean/full-data/hosted gap

- **metric_name 1: row 1**
- **metric_name 2: row 2**
- **metric_name 3: row 3**

12. standardized casewise results

- **model_id 1: mosaic_clean_oof**
  - metric_name: lesion_union
  - case_count: 220
  - mean_dice: 0.3367
  - empty_pred_count: 0
- **model_id 2: mosaic_clean_oof**
  - metric_name: pure_edema
  - case_count: 80
  - mean_dice: 0.0528
  - empty_pred_count: 64
- **model_id 3: mosaic_clean_oof**
  - metric_name: scar
  - case_count: 220
  - mean_dice: 0.3782
  - empty_pred_count: 0
- **model_id 4: nnunet_oof**
  - metric_name: lesion_union
  - case_count: 220
  - mean_dice: 0.5755
  - empty_pred_count: 3
- **model_id 5: nnunet_oof**
  - metric_name: pure_edema
  - case_count: 80
  - mean_dice: 0.4308
  - empty_pred_count: 0
- **model_id 6: nnunet_oof**
  - metric_name: scar
  - case_count: 220
  - mean_dice: 0.561
  - empty_pred_count: 3

13. scar failure taxonomy

Scar scientific brief

Scar evidence is dominated by label-5 lesion localization, small/multi-component false negatives, remote false positives and decoder capability loss. nnU-Net remains the strongest same-split anchor; Batch7 shows that making SRR modules visible or trainable does not guarantee useful final-mask authority. MMRD direct and distillation residual heads underperform nnU-Net, ARC exposes decoder-restoration risk, and PRISM W3 failed the outer gate. Useful carry-forward items are final-output-entry auditing, help/harm accounting, and pathology-specific candidate supervision. Forbidden repeats are decoder reset, prototype claims without isolation, module-present-only validation and leaving architecture blanks to execution.

14. pure-edema failure taxonomy

Pure-edema scientific brief

Pure edema is a T2-dependent label-4 problem and cannot borrow scar conclusions. The key evidence boundary is the 80 T2-present denominator; no-T2 cases must not become edema negatives. MoSAIC M0/M1 clean evidence is broad, but M2-M10 recipe decomposition is only six cases, and V3 feature probes still have many single-class edema folds. MMRD contributes a data hygiene rule more than a model win; Cascade shows tiny bounded correction; ARC and PRISM do not prove edema recovery. Future designs need an edema-specific mechanism with T2-aware supervision, center-stable feature evidence and an explicit pure-edema error budget.

15. frozen feature probes

V4 feature probe: `PASS_V4_PATIENT_LEVEL_REFOLD`；病例数 `80`；fold 数 `5`；outer 只读诊断，不参与 checkpoint、threshold 或后处理选择。

- **feature_source 1: VOLUME_CTRL**
  - task_id: P1_scar
  - passing_folds: 5
  - mean_AUROC: 0.5042
  - mean_AUPRC: 0.4927
- **feature_source 2: VOLUME_CTRL**
  - task_id: P2_scar_FN
  - passing_folds: 5
  - mean_AUROC: 0.507
  - mean_AUPRC: 0.4723
- **feature_source 3: VOLUME_CTRL**
  - task_id: P3_scar_FP
  - passing_folds: 5
  - mean_AUROC: 0.5125
  - mean_AUPRC: 0.4551
- **feature_source 4: VOLUME_CTRL**
  - task_id: P7_small_scar
  - passing_folds: 5
  - mean_AUROC: 0.6978
  - mean_AUPRC: 0.2891
- **feature_source 5: VOLUME_CTRL**
  - task_id: P8_boundary
  - passing_folds: 5
  - mean_AUROC: 0.5042
  - mean_AUPRC: 0.4927
- **feature_source 6: CENTER_CTRL**
  - task_id: P1_scar
  - passing_folds: 5
  - mean_AUROC: 0.5
  - mean_AUPRC: 0.4894
- **feature_source 7: CENTER_CTRL**
  - task_id: P2_scar_FN
  - passing_folds: 5
  - mean_AUROC: 0.5102
  - mean_AUPRC: 0.4707
- **feature_source 8: CENTER_CTRL**
  - task_id: P3_scar_FP
  - passing_folds: 5
  - mean_AUROC: 0.5029
  - mean_AUPRC: 0.4441
- **feature_source 9: CENTER_CTRL**
  - task_id: P7_small_scar
  - passing_folds: 5
  - mean_AUROC: 0.352
  - mean_AUPRC: 0.1239
- **feature_source 10: CENTER_CTRL**
  - task_id: P8_boundary
  - passing_folds: 5
  - mean_AUROC: 0.5
  - mean_AUPRC: 0.4894
- **feature_source 11: MODALITY_CTRL**
  - task_id: P1_scar
  - passing_folds: 5
  - mean_AUROC: 0.5
  - mean_AUPRC: 0.4894
- **feature_source 12: MODALITY_CTRL**
  - task_id: P2_scar_FN
  - passing_folds: 5

- mean_AUROC: 0.5
- mean_AUPRC: 0.4654

- **feature_source 1: VOLUME_CTRL**
  - task_id: P4_edema
  - passing_folds: 5
  - mean_AUROC: 0.5083
  - mean_AUPRC: 0.4983
- **feature_source 2: VOLUME_CTRL**
  - task_id: P5_edema_FN
  - passing_folds: 5
  - mean_AUROC: 0.5118
  - mean_AUPRC: 0.4793
- **feature_source 3: VOLUME_CTRL**
  - task_id: P6_edema_FP
  - passing_folds: 5
  - mean_AUROC: 0.5126
  - mean_AUPRC: 0.4789
- **feature_source 4: CENTER_CTRL**
  - task_id: P4_edema
  - passing_folds: 5
  - mean_AUROC: 0.5037
  - mean_AUPRC: 0.4941
- **feature_source 5: CENTER_CTRL**
  - task_id: P5_edema_FN
  - passing_folds: 5
  - mean_AUROC: 0.5079
  - mean_AUPRC: 0.4749
- **feature_source 6: CENTER_CTRL**
  - task_id: P6_edema_FP
  - passing_folds: 5
  - mean_AUROC: 0.5091
  - mean_AUPRC: 0.4735
- **feature_source 7: MODALITY_CTRL**
  - task_id: P4_edema
  - passing_folds: 5
  - mean_AUROC: 0.5
  - mean_AUPRC: 0.4922
- **feature_source 8: MODALITY_CTRL**
  - task_id: P5_edema_FN
  - passing_folds: 5
  - mean_AUROC: 0.5
  - mean_AUPRC: 0.4708
- **feature_source 9: MODALITY_CTRL**
  - task_id: P6_edema_FP
  - passing_folds: 5
  - mean_AUROC: 0.5
  - mean_AUPRC: 0.4688
- **feature_source 10: NNUNET_DECODER_L0**
  - task_id: P4_edema
  - passing_folds: 5
  - mean_AUROC: 0.7597
  - mean_AUPRC: 0.6484
- **feature_source 11: NNUNET_DECODER_L0**
  - task_id: P5_edema_FN
  - passing_folds: 5
  - mean_AUROC: 0.7522
  - mean_AUPRC: 0.5759
- **feature_source 12: NNUNET_DECODER_L0**
  - task_id: P6_edema_FP
  - passing_folds: 5
  - mean_AUROC: 0.7359
  - mean_AUPRC: 0.5733

16. alignment

Alignment conclusion

V3 contains real complete-trimodal alignment measurements, so alignment is no longer only a plan. V4 recomputes bootstrap correlation intervals, center-adjusted slopes and high/low shift subgroups from the bound casewise rows. The current evidence supports alignment as an optional diagnostic or safety module, not the primary Deep Research mechanism.

- **alignment_metric 1: row 1**
  - bootstrap_ci_low: -0.4797
  - bootstrap_ci_high: 0.1597
- **alignment_metric 2: row 2**
  - bootstrap_ci_low: -0.228
  - bootstrap_ci_high: 0.598
- **alignment_metric 3: row 3**
  - bootstrap_ci_low: -0.5388
  - bootstrap_ci_high: 0.1861
- **alignment_metric 4: row 4**
  - bootstrap_ci_low: -0.6215
  - bootstrap_ci_high: -0.002
- **alignment_metric 5: row 5**
  - bootstrap_ci_low: 0.1454
  - bootstrap_ci_high: 0.6757
- **alignment_metric 6: row 6**
  - bootstrap_ci_low: -0.4872
  - bootstrap_ci_high: 0.0695
- **alignment_metric 7: row 7**
  - bootstrap_ci_low: -0.0935
  - bootstrap_ci_high: 0.5721
- **alignment_metric 8: row 8**
  - bootstrap_ci_low: -0.6429
  - bootstrap_ci_high: -0.0381
- **alignment_metric 9: row 9**
  - bootstrap_ci_low: -0.1578
  - bootstrap_ci_high: 0.4587
- **alignment_metric 10: row 10**
  - bootstrap_ci_low: -0.0285
  - bootstrap_ci_high: 0.6149
- **alignment_metric 11: row 11**
  - bootstrap_ci_low: -0.7464
  - bootstrap_ci_high: -0.228
- **alignment_metric 12: row 12**
  - bootstrap_ci_low: -0.0422
  - bootstrap_ci_high: 0.5437

17. help/harm 和 oracle

- **case_id 1: Case1002**
  - metric_name: scar
  - best_case_model: nnunet_oof
  - case_oracle_dice: 0.6167
  - voxel_tp_oracle_dice: 0.7338
- **case_id 2: Case1002**
  - metric_name: lesion_union
  - best_case_model: nnunet_oof
  - case_oracle_dice: 0.6167
  - voxel_tp_oracle_dice: 0.8947
- **case_id 3: Case1007**
  - metric_name: scar
  - best_case_model: nnunet_oof
  - case_oracle_dice: 0.5811
  - voxel_tp_oracle_dice: 0.912

- **case_id 4: Case1007**
  - metric_name: lesion_union
  - best_case_model: nnunet_oof
  - case_oracle_dice: 0.5811
  - voxel_tp_oracle_dice: 0.9487
- **case_id 5: Case1009**
  - metric_name: scar
  - best_case_model: nnunet_oof
  - case_oracle_dice: 0.6049
  - voxel_tp_oracle_dice: 0.7778
- **case_id 6: Case1009**
  - metric_name: lesion_union
  - best_case_model: nnunet_oof
  - case_oracle_dice: 0.6049
  - voxel_tp_oracle_dice: 0.8556
- **case_id 7: Case1010**
  - metric_name: scar
  - best_case_model: nnunet_oof
  - case_oracle_dice: 0.4828
  - voxel_tp_oracle_dice: 0.789
- **case_id 8: Case1010**
  - metric_name: lesion_union
  - best_case_model: nnunet_oof
  - case_oracle_dice: 0.4828
  - voxel_tp_oracle_dice: 0.9466
- **case_id 9: Case1021**
  - metric_name: scar
  - best_case_model: nnunet_oof
  - case_oracle_dice: 0.5699
  - voxel_tp_oracle_dice: 0.8076
- **case_id 10: Case1021**
  - metric_name: lesion_union
  - best_case_model: nnunet_oof
  - case_oracle_dice: 0.5699
  - voxel_tp_oracle_dice: 0.8911
- **case_id 11: Case1023**
  - metric_name: scar
  - best_case_model: nnunet_oof
  - case_oracle_dice: 0.6466
  - voxel_tp_oracle_dice: 0.8091
- **case_id 12: Case1023**
  - metric_name: lesion_union
  - best_case_model: nnunet_oof
  - case_oracle_dice: 0.6466
  - voxel_tp_oracle_dice: 0.9032

18. large-gain error budget

Large-gain conclusion

Current local evidence argues against getting approximately 0.1 Dice from selector/recipe reuse alone. Scar has a small case-oracle gain over nnU-Net; pure edema has an even smaller case-oracle gain but a larger non-deployable voxel oracle, indicating the need for an external mechanism rather than another weak anchor correction.

- **pathology 1: scar**
  - case_oracle_bound: 0.022
  - voxel_oracle_bound: 0.2375
  - selector_bound: 0.8269
  - conclusion: ONLY_MODEST_GAIN
- **pathology 2: pure_edema**
  - case_oracle_bound: 0.0023
  - voxel_oracle_bound: 0.173
  - selector_bound: 0
  - conclusion: ONLY_MODEST_GAIN

19. component survival

Component survival report

Directly reusable experience:

- Retain reliable-label/no-T2 masking as a data rule for edema.
- Retain help/harm, remote-FP and final-output-entry accounting as safety gates.

Forbidden repeats:

- Decoder reset or encoder-only inheritance presented as a full decoder.
- Module-present or gradient-nonzero evidence presented as final-output mechanism success.
- no-T2 edema misuse as negative supervision.
- Weak bounded correction around an anchor without an error selector.
- Prototype experiment conclusions without isolated control inputs.
- Architecture blanks delegated to Codex/controller during execution.

20. visual atlas

独立 atlas 作为单独 PDF 保存，主 PDF 只嵌入 contact sheet，避免再引入右侧 panel 裁切。

- **field 1: atlas_pdf**
  - value: v4_atlas_pages_a3_landscape.pdf
- **field 2: atlas_page_count**
  - value: 40
- **field 3: bbox_validation**
  - value: v4_atlas_pdf_bbox_validation.csv
- **field 4: bbox_status**
  - value: all page bbox rows PASS

21. Deep Research constraints

DEEP RESEARCH MODEL DESIGN INPUT 20260730 V4

This file is a design-input constraint packet, not a new model design. It records what the next Deep Research design may and may not use from the failure-forensics evidence.

Current readiness

- scientific_evidence_status: `SUFFICIENT`
- deep_research_readiness: `READY`
- current_model_status: `FAILED_GATE`

Current strong baselines and data truth

- Total MyoPS training cases: 220.
- T2-present official pure-edema denominator: 80.
- C0-present cases: 104.
- nnU-Net clean and MoSAIC clean remain baseline/context evidence; PRISM W3 failed the outer gate and must not be promoted.

Historical experience allowed as inputs

- Batch7: `pathology-specific proposal/refiner` -> RETEST_WITH_DIFFERENT_IMPLEMENTATION; use only with precondition `isolated patient-held-out causal ablation`.
- MMRD: `reliable-label no-T2 mask` -> RETAIN_AS_DATA_RULE; use only with precondition `T2-present split accounting`.
- ARC: `single encoder with explicit modality gates` -> RETAIN_AS_SAFETY_RULE; use only with precondition `restore decoder capability before claiming gain`.

Must not repeat

- Do not stack multiple complete backbones as the central method.
- Do not use encoder-only inheritance or decoder reset as a full decoder.
- Do not let nnU-Net or MoSAIC be the only final authority.
- Do not use identical scar and edema heads.
- Do not treat no-T2 cases as pure-edema negatives.
- Do not claim prototype value from unisolated control tensors.
- Do not use weak correction around an anchor without an error selector.
- Do not leave proposal/refiner wiring or component definitions for Codex/controller to invent.
- Do not treat gradient/nonzero-delta validators as causal mechanism proof.

Large-gain boundary

- scar: case oracle=0.02195407548910211, voxel oracle=0.23751872769841142, conclusion=`ONLY_MODEST_GAIN`.
- pure_edema: case oracle=0.002292654276233319, voxel oracle=0.17295548404011052, conclusion=`ONLY_MODEST_GAIN`.

Open requirements before READY

22. current conclusions

科学证据状态为 `SUFFICIENT`，Deep Research readiness 为 `READY`。这只表示下一代模型设计已有足够约束输入；当前 CARE 模型仍为 `FAILED_GATE`，没有新架构训练、没有 validation upload、没有 Docker upload、没有 hosted metric claim。

23. unresolved questions

- [DR-001] Small-lesion scar segmentation beyond nnU-Net requires which evidence standard?
- [DR-002] Can clean MoSAIC recipe gains be separated from full-data target-domain advantage?
- [DR-003] Do frozen encoder features contain patient-held-out scar FN/FP separability?
- [DR-004] When does cine temporal information improve pathology segmentation over ED-only?

附录 A. validator 和 claim ledger

- **requirement 1: Batch7 no longer empty**
  - passed: True
  - evidence: v4 current artifact
- **requirement 2: MMRD direct/distillation casewise bound**
  - passed: True
  - evidence: v4 current artifact
- **requirement 3: Cascade prototype semantics audited**
  - passed: True
  - evidence: v4 current artifact
- **requirement 4: ARC blueprint-code-runtime completed**
  - passed: True
  - evidence: v4 current artifact
- **requirement 5: MoSAIC population sufficient**
  - passed: True
  - evidence: mosaic_ recipe_ decomposition_ receipt.json; v4_ mosaic_ recipe_ population_ audit.json; v4_ mosaic_ m0_ m10_ casewise.csv
- **requirement 6: Pure-edema feature probe completed**
  - passed: True
  - evidence: v4_ feature_ probe_ receipt.json; v4_ feature_ probe_ leakage_ audit.json; v4_ feature_ probe_ fold_ results.csv; v4_ feature_ probe_ edema_ summary.csv
- **requirement 7: Scar/edema briefs independent**
  - passed: True
  - evidence: v4 current artifact
- **requirement 8: Large-gain error budget completed**
  - passed: True
  - evidence: v4_ large_ gain_ error_ budget.csv; v4_ large_ gain_ bounds.csv
- **requirement 9: Alignment completed**
  - passed: True
  - evidence: alignment_ v2_ receipt.json; v4_ alignment_ failure_ correlation.csv; v4_ alignment_ subgroup_ results.csv
- **requirement 10: Visual atlas no clipping**
  - passed: True
  - evidence: v4_ atlas_ pages_ a3_ landscape.pdf; v4_ atlas_ pdf_ bbox_ validation.csv
- **requirement 11: State contradictions removed**
  - passed: True
  - evidence: v4_ state_ semantics_ contract.json; v4_ v3_ state_ contradiction.csv; v4_ superseded_ statement_ manifest.csv
- **requirement 12: Claim ledger complete**
  - passed: True
  - evidence: evidence_ claim_ ledger.csv plus V4 component, feature, MoSAIC, alignment, large- gain and state ledgers

附录 B. Slurm accounting

- **job_id 1: 61220581.51**
  - name: bash
  - state: COMPLETED
  - exit_code: 0:0
  - submit: 2026-07-30T10:57:18
  - end: 2026-07-30T10:59:39
- **job_id 2: 61220581.52**
  - name: bash
  - state: COMPLETED
  - exit_code: 0:0
  - submit: 2026-07-30T11:02:22
  - end: 2026-07-30T11:04:32
- **job_id 3: 61220581.53**
  - name: bash
  - state: FAILED
  - exit_code: 1:0
  - submit: 2026-07-30T11:14:13

- end: 2026-07-30T11:16:46
- **job_id 4: 61220581.54**
  - name: bash
  - state: CANCELLED by 397557
  - exit_code: 0:9
  - submit: 2026-07-30T11:17:16
  - end: 2026-07-30T11:22:07
- **job_id 5: 61220581.55**
  - name: bash
  - state: FAILED
  - exit_code: 1:0
  - submit: 2026-07-30T11:23:16
  - end: 2026-07-30T11:23:43
- **job_id 6: 61220581.56**
  - name: bash
  - state: CANCELLED by 397557
  - exit_code: 0:9
  - submit: 2026-07-30T11:24:07
  - end: 2026-07-30T11:27:06
- **job_id 7: 61220581.57**
  - name: bash
  - state: FAILED
  - exit_code: 2:0
  - submit: 2026-07-30T11:28:11
  - end: 2026-07-30T11:31:05
- **job_id 8: 61220581.58**
  - name: bash
  - state: COMPLETED
  - exit_code: 0:0
  - submit: 2026-07-30T11:32:15
  - end: 2026-07-30T11:35:34
- **job_id 9: 61376439**
  - name: V4Mosaic80
  - state: CANCELLED by 397557
  - exit_code: 0:0
  - submit: 2026-07-30T10:38:26
  - end: 2026-07-30T11:08:54

附录 C. provenance blocks

- **claim_id: E-DATA-001**
  - source_path: data_case_manifest.csv
  - confidence: MODERATE
  - notes: geometry round-trip incomplete
- **claim_id: E-METRIC-001**
  - source_path: reference_metric_known_bad_report.json
  - confidence: HIGH
  - notes: synthetic fixtures only
- **claim_id: E-MOSAIC-001**
  - source_path: mosaic_ablation_contract.json
  - confidence: HIGH
  - notes: contract boundary
- **claim_id: E-PRISM-001**
  - source_path: decoder_reset_diagnostic_report.md
  - confidence: LOW
  - notes: diagnostics not run
- **claim_id: E-GAP-005**
  - source_path: strict_validator_report.json
  - confidence: HIGH
  - notes: validator will fail until completed
- **claim_id: E-GAP-006**
  - source_path: strict_validator_report.json

- confidence: HIGH
- notes: validator will fail until completed

- **claim_id: E-GAP-007**
  - source_path: strict_validator_report.json
  - confidence: HIGH
  - notes: validator will fail until completed
- **claim_id: E-GAP-008**
  - source_path: strict_validator_report.json
  - confidence: HIGH
  - notes: validator will fail until completed
- **claim_id: E-GAP-009**
  - source_path: strict_validator_report.json
  - confidence: HIGH
  - notes: validator will fail until completed
- **claim_id: E-GAP-010**
  - source_path: strict_validator_report.json
  - confidence: HIGH
  - notes: validator will fail until completed
- **claim_id: E-GAP-011**
  - source_path: strict_validator_report.json
  - confidence: HIGH
  - notes: validator will fail until completed
- **claim_id: E-GAP-012**
  - source_path: strict_validator_report.json
  - confidence: HIGH
  - notes: validator will fail until completed